Chapter 2 - The Real Number System

Problem Set # 1

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1. If $a, b \in \mathbb{R}$, prove the following:
   (a) If $a + b = 0$, then $b = -a$
   (b) $-(-a) = a$
   (c) $(-1)a = -a$
   (d) $(-1)(-1) = 1$
2. Prove that if $a, b \in \mathbb{R}$, then
   (a) $-(a + b) = (-a) + (-b)$
   (b) $(-a) \cdot (-b) = a \cdot b$
   (c) $1/(-a) = -(1/a)$
   (d) $(a/b) = (-a)/b$ if $b \neq 0$
3. Solve the following equations, justifying each step by referring to an appropriate property or theorem.
   (a) $2x + 5 = 8$
   (b) $x^2 = 2x$
   (c) $x^2 - 1 = 3$
   (d) $(x - 1)(x + 2) = 0$
4. If $a \in \mathbb{R}$ satisfies $a \cdot a = a$, prove that either $a = 0$ or $a = 1$.
5. If $a \neq 0$ and $b \neq 0$, show that $1/(ab) = (1/a)(1/b)$.
6. Show that there does not exist a rational number s such that $s^2 = 6$.
7. Show that there does not exist a rational number t such that $t^2 = 3$.

8. (a) Show that if x, y are rational numbers, then $x + y$ and xy are rational numbers.

   (b) Prove that if x is a rational number and y is an irrational number, then $x+y$ is an irrational number. If, in addition, $x \neq 0$, then show that xy is an irrational number.

9. Let $K := \{s + t\sqrt{2} : s, t \in \mathbb{Q}\}$. Show that K satisfies the following:

   (a) If $x_1, x_2 \in K$, then $x_1 + x_2 \in K$ and $x_1 x_2 \in K$.

   (b) If $x \neq 0$ and $x \in K$, then $1/x \in K$.

   (Thus the set K is a *subfield* of $\mathbb{R}$. With the order inherited from $\mathbb{R}$, the set K is an ordered field that lies between $\mathbb{Q}$ and $\mathbb{R}$.)

10. (a) If $a < b$ and $c \leq d$, prove that $a + c < b + d$.

    (b) If $0 < a < b$ and $0 \leq c \leq d$, prove that $0 \leq ac \leq bd$.

11. (a) Show that if $a > 0$, then $1/a > 0$ and $1/(1/a) = a$.

    (b) Show that if $a < b$, then $a < \frac{1}{2}(a + b) < b$.

12. Let a, b, c, d be numbers satisfying $0 < a < b$ and $c < d < 0$. Give an example where $ac < bd$, and another where $bd < ac$.

13. If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ if and only if $a = 0$ and $b = 0$.

14. If $0 \leq a < b$, show that $a^2 \leq ab < b^2$. Show by example that it does **not** follow that $\sqrt{a} \leq \sqrt{b}$.

15. Prove that for $0 < a < 1$, (a) $a < \sqrt{ab} < b$, and (b) $1/b < 1/a$.

16. Find all real numbers x that satisfy the following inequalities.

    (a) $x^2 > 3x + 4$

    (b) $1 < x^2 < 4$

    (c) $1/x < x$

    (d) $1/x < x^2$

17. Prove: If $a \in \mathbb{R}$ is such that $0 \leq a \leq \varepsilon$ for every $\varepsilon > 0$, then $a = 0$.

18. Let $a, b \in \mathbb{R}$, and suppose that for every $\varepsilon > 0$ we have $a \leq b + \varepsilon$. Show that $a \leq b$.

19. Prove that $\left[\frac{1}{2}(a + b)\right]^2 \leq \frac{1}{2}(a^2 + b^2)$ for all $a, b \in \mathbb{R}$ and also that the equality holds if and only if $a = b$.

20. (a) If $0 < c < 1$, show that $0 < c^2 < c < 1$.

    (b) If $1 < c$, show that $1 < c < c^2$.

21. (a) Prove there is no $n \in \mathbb{Z}^+$ such that $0 < n < 1$. (Use the Well-Ordering Property of $\mathbb{Z}^+$).

    (b) Prove that no natural number can be both even and odd.

22. (a) If $c > 1$, show that $c^n \geq c$ for all $n \in \mathbb{Z}^+$, and that $c^n > c$ for $n > 1$.

    (b) If $0 < c < 1$, show that $c^n \leq c$ for all $n \in \mathbb{Z}^+$, and that $c^n < c$ for $n > 1$.

23. If $a > 0$, $b > 0$, and $n \in \mathbb{Z}^+$, show that $a < b$ if and only if $a^n < b^n$. [*Hint: Use Mathematical Induction.*]

24. (a) If $c > 1$ and $m, n \in \mathbb{Z}^+$, show that $c^m > c^n$ if and only if $m > n$.

    (b) If $0 < c < 1$ and $m, n \in \mathbb{Z}^+$, show that $c^m < c^n$ if and only if $m > n$.

25. Assuming the existence of roots, show that if $c > 1$, then $c^{1/m} < c^{1/n}$ if and only if $m > n$.

26. Use Mathematical Induction to show that if $a \in \mathbb{R}$ and $m, n \in \mathbb{Z}^+$, then $a^{m+n} = a^m a^n$ and $(a^m)^n = a^{mn}$.

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"Success is the sum of small efforts, repeated day in and day out." - Robert Collier -